Kesavan Mohana Sundaram | Curriculum Vitae

✉ km2@huskers.unl.edu • arXiv • kesavan-ms.github.io
Office: 326 Avery Hall, University of Nebraska–Lincoln

Education

University of Nebraska–Lincoln — **Lincoln, Nebraska**
PhD Candidate in Mathematics — *2023–2028*
Advisors: Eloísa Grifo and Jack Jeffries

University of Nebraska–Lincoln — **Lincoln, Nebraska**
M.S. in Mathematics — *2023–2025*

Indian Institute of Technology Madras — **Chennai, India**
M.Sc. in Mathematics — *2021–2023*

Ramakrishna Mission Vivekananda College — **Chennai, India**
B.Sc. in Mathematics — *2016–2019*

Planned Research Visit

SLMath (formerly known as MSRI) — **Berkeley, California**
AxIOM program member — *Jan–Feb 2027*
Topic: Commutative Algebra, Representation Theory, and other Interactions

Research

I am interested in commutative algebra and homological algebra, and their interactions with algebraic geometry. In particular: arithmetic rank, cohomological support varieties, local cohomology, modules of finite length and finite projective dimension, and symbolic powers.

Published papers

1. *The arithmetic rank of the residual intersections of a complete intersection ideal*,
   with Manav Batavia, Taylor Murray, and Vaibhav Pandey
   International Mathematics Research Notices (2026), no. 10, rnag091.
   DOI: 10.1093/imrn/rnag091 arXiv

Preprints

2. *Lifting systems for finite length modules*,
   with Benjamin Katz, Nawaj KC, Andrew J. Soto Levins, and Ryan Watson arXiv
3. *Cohomological support varieties for monomial ideals*,
   with Kara Fagerstrom, Julianne Faur, Benjamin Katz, Stephen Stern, and Ryan Watson arXiv
4. *The arithmetic rank of nullcones of classical invariant rings*,
   with Manav Batavia and Aryaman Maithani arXiv
5. *Symbolic powers and asymptotic invariants of invariant ideals in determinantal rings*,
   with Sankhaneel Bisui, Aryaman Maithani, Alexandra Seceleanu, and Stephen Stern

Honors and Awards

- UNL Chancellor's Fellowship, 2023–2025.
- IIT Madras Merit Scholarship, 2022–2023.

Mentoring

Directed Reading Program

Positive characteristic methods in commutative algebra *Fall 2026*
Undergraduate mentee: Maxim Karpuk (co-advising with Robert Ireland)
Current reading: §1 in Frobenius notes by Jeffries; §1–2 in *Beginning in Algebraic Geometry* by Clader–Ross.

Representation theory of finite groups *Spring 2026*
Undergraduate mentee: Eric Woods; currently a graduate student at UNL
Read through §8 in *Groups and their Representations* by Karen Smith.

Service and Organization

Mathematics Department AI Task Force *Fall 2026*
Committee member, University of Nebraska–Lincoln

Directed Reading Program *Fall 2025–Present*
Co-organizer, University of Nebraska–Lincoln; Website
Undergraduate reading projects with graduate mentors and end-of-semester presentations.

Commutative Algebra and Adjacent Book-club Session *Spring 2026*
Co-organizer, University of Nebraska–Lincoln; Website
Collaborative reading and problem sessions on integral closure using Huneke–Swanson textbook.

Cohomology Club *Fall 2025*
Co-organizer, University of Nebraska–Lincoln; Website
Collaborative reading group on cohomology using Hatcher's *Algebraic Topology*.

Virtual Commutative Algebra and Algebraic Geometry Seminar *2024–2026*
Co-organizer; Website

Commutative Algebra Reading Seminar *Spring 2024, Summer 2024–2026*
Co-organizer, University of Nebraska–Lincoln; Website

Commutative Algebra Student Seminar *June–July 2023*
Co-organizer, Indian Institute of Technology Madras

FORAYS 2023 *February 2023*
Co-organizer, Indian Institute of Technology Madras; Website
Mathematics outreach activities for school and undergraduate students.

Teaching Experience

University of Nebraska–Lincoln

Instructor of record

Math 103 – College Algebra and Trigonometry *Fall 2025*
Math 102 – Trigonometry *Fall 2024, Spring 2025*

Graduate Teaching Assistant

Math 208 – Calculus III *Spring 2026, Fall 2026*
Math 107 – Calculus II *Spring 2024*
Math 106 – Calculus I *Fall 2023, Summer 2024*

Grader for courses

Math 905 – Commutative Algebra I *Fall 2026*
Math 818 – Introduction to Modern Algebra II *Spring 2026*
Math 309 – Introduction to Proofs *Fall 2025*

Talks

Upcoming

December 2026: CMS Winter Meeting, Montréal, Canada.
April 2027: AMS Spring 2027 Central Sectional Meeting, Iowa City.

Invited talks

AMS Spring Central Sectional Meeting, North Dakota State University
The arithmetic rank of residual intersections of a complete intersection ideal *April 2026*

Graduate Student Seminar, University of Tennessee
The arithmetic rank of residual intersections of a complete intersection ideal *April 2026*

Mathematics Special Seminar, IIT Madras
The arithmetic rank of residual intersections of a complete intersection ideal *December 2025*

Student Algebra Seminar, Purdue University, virtual
Lifting systems for finite length modules *November 2025*

Colegio de Matemáticos del Perú, virtual
The arithmetic rank of residual intersections of a complete intersection ideal *October 2025*

Seminar talks

Virtual Commutative Algebra and Algebraic Geometry Seminar
Cohomological support varieties for monomial ideals *September 2026*

UNL RTG Learning Seminar
Are quadratic Gorenstein algebras always Koszul? *September 2026*

UNL Commutative Algebra Reading Seminar
On the annihilators of local cohomology modules *August 2026*

UNL Commutative Algebra Reading Seminar
Vanishing theorems in local cohomology *July 2026*

UNL Graduate Student Seminar
Differential operators on polynomial rings *March 2026*

UNL RTG Learning Seminar
Regularity of modules over a Koszul algebra *March 2026*

UNL Commutative Algebra Reading Seminar
A topological approach to compute the arithmetic rank *March 2026*

UNL Cohomology Club
An introduction to Čech cohomology *December 2025*

UNL Graduate Student Seminar
Algebras with a Straightening Law *October 2025*

UNL Commutative Algebra Seminar
The arithmetic rank of residual intersections of a complete intersection ideal *October 2025*

Virtual Commutative Algebra and Algebraic Geometry Seminar
The derived category of a DG algebra *October 2025*

UNL Commutative Algebra Reading Seminar
A stratification by F-pure threshold *September 2025*

UNL Commutative Algebra Reading Seminar
Measuring singularities with Frobenius *July 2025*

UNL Commutative Algebra Reading Seminar
Arithmetic rank and cohomological dimension in commutative algebra *May 2025*

UNL Graduate Student Seminar
DG algebra resolutions *March 2025*

UNL Big CMinar
Direct summand conjecture in mixed characteristic *March 2025*

Virtual Commutative Algebra and Algebraic Geometry Seminar
Kähler differentials *Fall 2024*

UNL Commutative Algebra Reading Seminar
A Zariski–Nagata theorem for smooth $\mathbb{Z}$-algebras *July 2024*

UNL Commutative Algebra Reading Seminar
An introduction to symbolic powers *June 2024*

UNL Graduate Student Seminar
Powers of ideals in commutative algebra *April 2024*

UNL Commutative Algebra Reading Seminar
Adams operations in commutative algebra *February 2024*

UNL Graduate Student Seminar
An introduction to Schur polynomials *September 2023*

IIT Madras Commutative Algebra Student Seminar
An introduction to homological algebra *June 2023*

Conferences and Programs Attended

New Homological Methods in Toric Geometry, Oberwolfach, Germany *October 2026*
URiCA (Upcoming Researchers in Commutative Algebra), Lincoln, Nebraska *August 2026*
Singularities in Commutative Algebra through Cohomological Methods, SLMath *July 2026*
Pan-American School in Commutative Algebra 2.0, CIMAT, Guanajuato, Mexico *May 2026*
A Celebration of Golod's Mathematical Legacy, Coventry, United Kingdom *April 2026*
AMS Spring Sectional Meeting, North Dakota State University, Fargo, North Dakota *April 2026*
CA+, Ames, Iowa *March 2026*
PL – Commutative Algebra and Modular Representation Theory, Chennai, India *December 2025*
KUMUNU, Columbia, Missouri *October 2025*
AMS Fall Central Sectional Meeting, St. Louis, Missouri *October 2025*
Summer CAMP 2.0 (Commutative Algebra Market Preparedness), Lincoln, Nebraska *August 2025*
Singularities in Algebra and Geometry, Guanajuato, Mexico *June 2025*
An Introduction to Recent Trends in Commutative Algebra, Toronto, Canada *June 2025*
URiCA (Upcoming Researchers in Commutative Algebra), Lincoln, Nebraska *May 2025*
Rayfest, Lincoln, Nebraska *April 2025*
Commutative Algebra in Mixed Characteristic, Lincoln, Nebraska *March 2025*
AIS – Cohomology of Commutative Algebras, Chennai, India *December 2024*
KUMUNU, Columbia, Missouri *September 2024*
BRIDGES, Salt Lake City, Utah *July 2024*
URiCA (Upcoming Researchers in Commutative Algebra), Lincoln, Nebraska *May 2024*
Recent Developments in Commutative Algebra, MSRI/SLMath, Berkeley, California *April 2024*
TAGS: Syzygies over Texas, College Station, Texas *April 2024*
Introductory Workshop: Commutative Algebra, MSRI/SLMath, Berkeley, California *January 2024*
Connections Workshop: Commutative Algebra, MSRI/SLMath, Berkeley, California *January 2024*
GEMS, Minneapolis, Minnesota *November 2023*
AMS Sectional Meeting, Omaha, Nebraska *October 2023*
KUMUNU, Columbia, Missouri *September 2023*
BrianFest, Lincoln, Nebraska *August 2023*
Summer CAMP (Commutative Algebra Market Preparedness), Lincoln, Nebraska *August 2023*